GOVERNMENT ENTERPRISE SPECIFICATION

14 Sovereign Citizen Identity Architecture

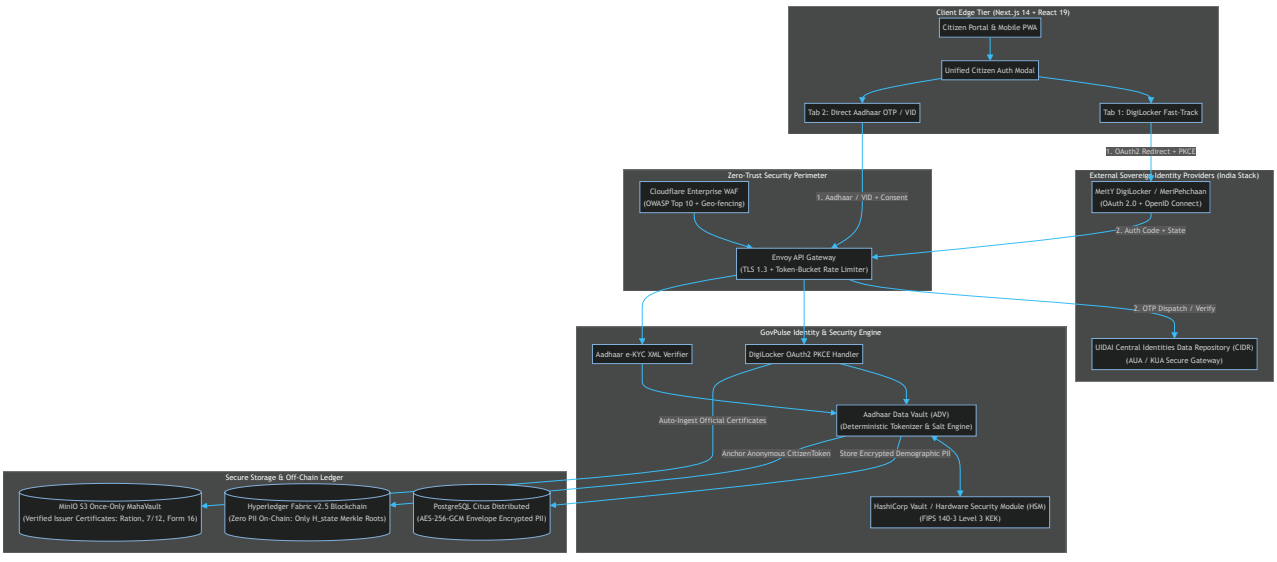
Unified Hybrid DigiLocker OAuth 2.0 & Aadhaar e-KYC • Aadhaar Data Vault • DPDP Act 2023 • Version 1.0

1. Architectural Decision Record (ADR-009)

Decision: Adopt **Option D (Unified Hybrid Architecture)** combining **MeitY DigiLocker OAuth 2.0 / MeriPehchaan** and **UIDAI Direct Aadhaar OTP e-KYC** with an isolated **Aadhaar Data Vault (ADV)**.

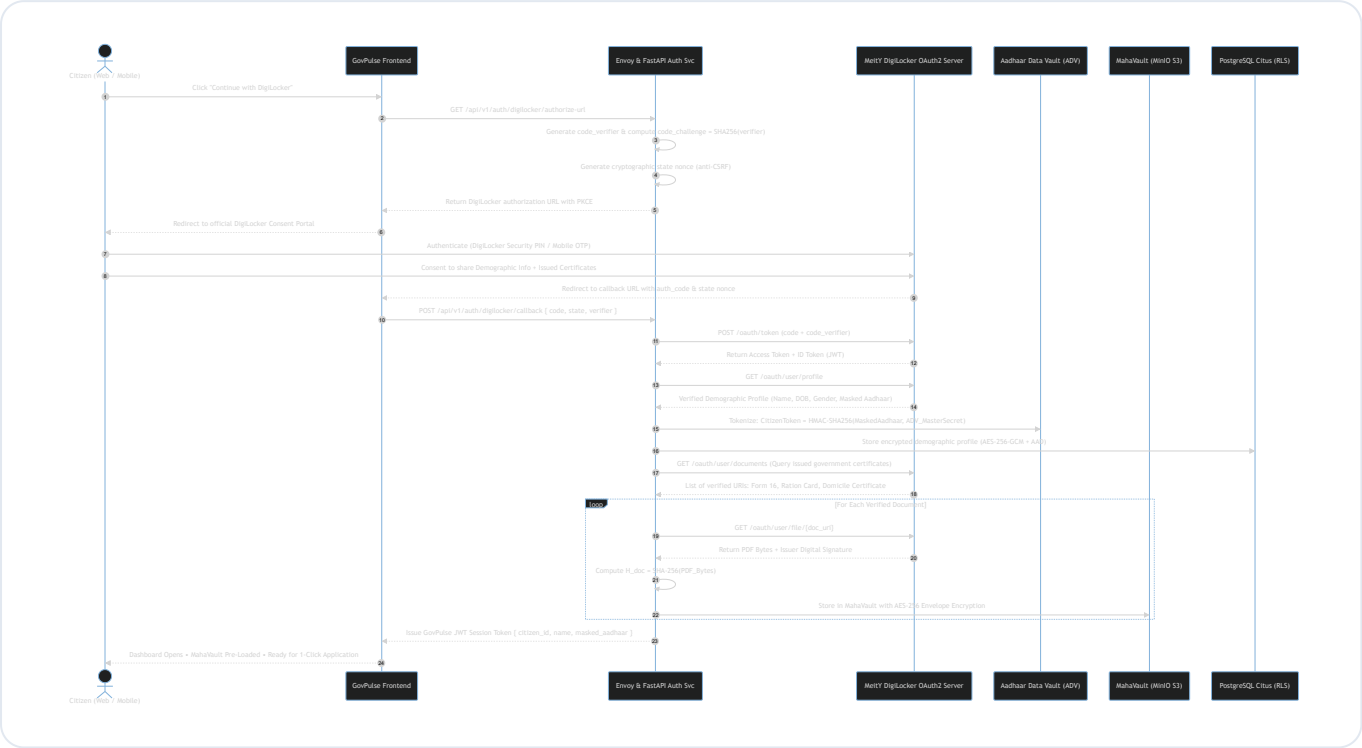
Rationale: Solves India's digital divide by providing 100% coverage (Aadhaar holders) while delivering 1-click automated document imports for DigiLocker holders. Conforms fully with the Supreme Court Puttaswamy judgment, UIDAI ADV regulations, and the India Digital Personal Data Protection (DPDP) Act 2023.

2. High-Level System Topology

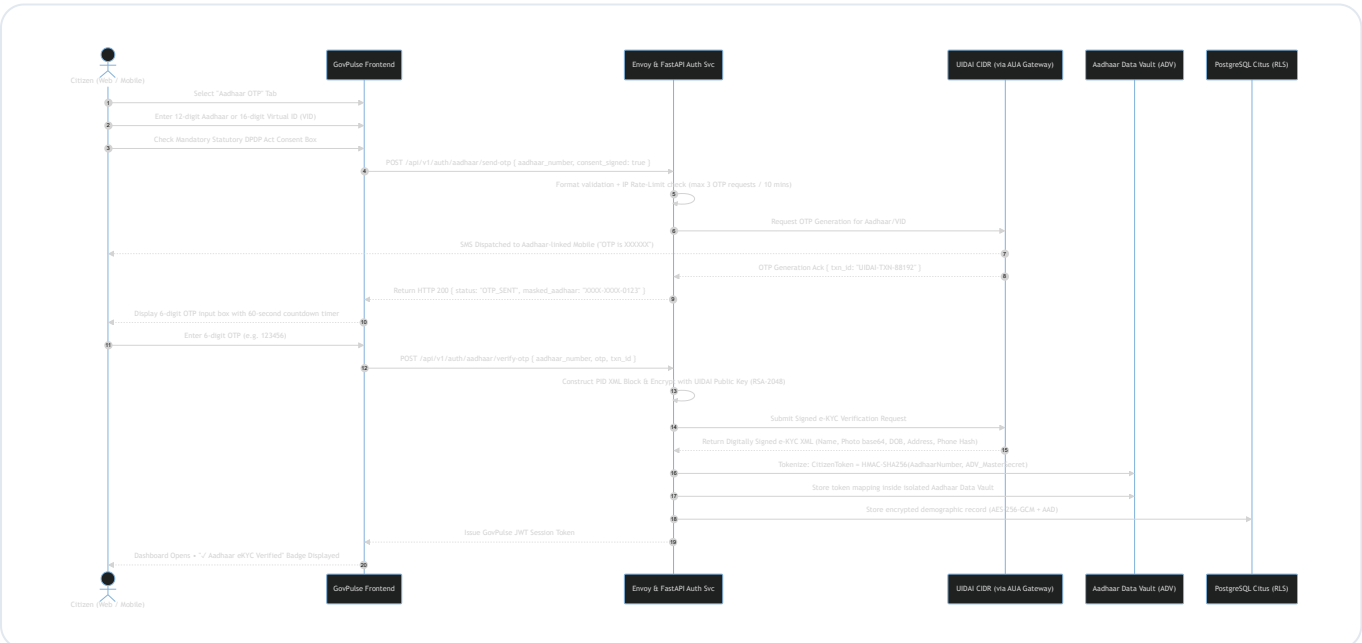


3. End-to-End Sequence Flows

Path 1: DigiLocker Fast-Track with Automatic MahaVault Synchronization



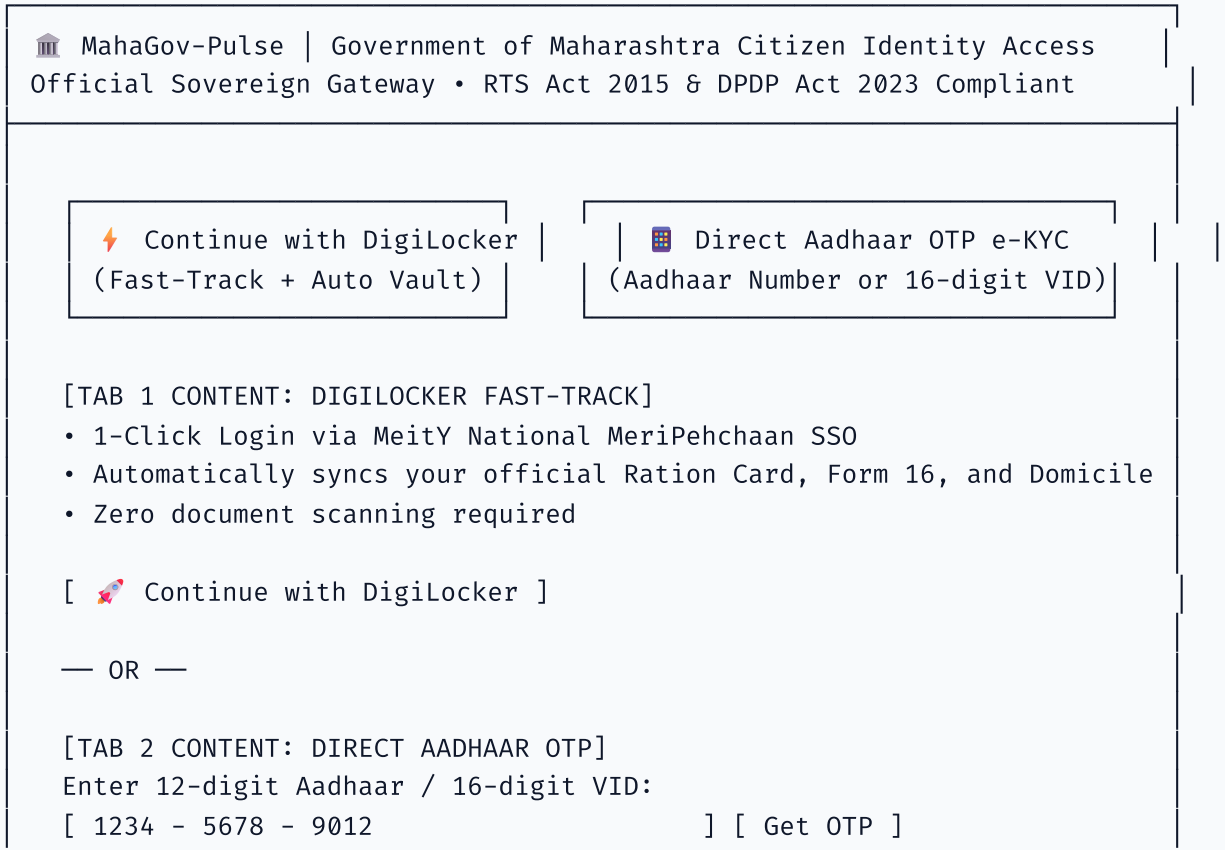
Path 2: Direct Aadhaar OTP e-KYC via UIDAI Gateway



4. Four Pillars of Identity Security & Data Protection

SECURITY PILLAR	ARCHITECTURAL IMPLEMENTATION	REGULATORY STANDARD
1. Aadhaar Data Vault (ADV)	The 12-digit raw Aadhaar is never saved in databases or logs. Deterministically converted to `CitizenToken = HMAC-SHA256(Aadhaar, HSM_Key)`.	UIDAI Circular F.No. 11020/205/2017
2. Zero PII On-Chain	Hyperledger Fabric records zero cleartext names, photos, or Aadhaar numbers. Only `H_state = SHA256( ... )` is committed.	DPDP Act 2023 §8 & GDPR Art. 17 (Right to Erasure)
3. Mandatory 8-Digit Masking	UI, APIs, and PDF certificates only expose `XXXX-XXXX-1234`. First 8 digits permanently masked.	Aadhaar Act 2016 §29 & RBI Guidelines
4. AES-256-GCM Envelope Encryption	Demographic PII in PostgreSQL Citus encrypted using per-user DEKs wrapped by HashiCorp Vault / HSM KEK.	FIPS 140-3 Level 3 & ISO/IEC 27001

6. Frontend Citizen Authentication Experience Blueprint



```
┌──────────────────────────────────────────────────────────────────┐
│                                                                  │
│   Enter 6-digit OTP sent to registered mobile:                   │
│   [ •  •  •  •  •  • ] (Resend in 45s)                           │
│                                                                  │
│   [  Verify & Secure Sign-In ]                                   │
│                                                                  │
│   ☑ Bilingual Statutory Consent:                                 │
│     "I hereby consent to provide my Aadhaar/DigiLocker credentials for
│      statutory certificate issuance under the Maharashtra RTS Act 2015
│      and the Digital Personal Data Protection (DPDP) Act 2023."
└──────────────────────────────────────────────────────────────────┘
```

7. Operational REST API Specifications

1. Generate DigiLocker Authorize URL

`GET /api/v1/auth/digilocker/authorize-url`


```
{
  "authorize_url": "https://digilocker.meripehchaan.gov.in/public/oauth2/1/authorize?
  "state": "a1b2c3d4e5f6",
  "code_verifier": "dBjftJeZ4CVP-mB92K27uhbUJU1p1r_wW1gFWFOEjXk"
}
```


2. Handle DigiLocker OAuth Callback

`POST /api/v1/auth/digilocker/callback`


```
{
  "access_token": "eyJhbGciOi ... ",
  "token_type": "bearer",
  "user_id": "CIT-MH-88190",
  "full_name_en": "Rahul Suresh Patil",
  "masked_aadhaar": "XXXX-XXXX-0123",
  "district": "Pune",
  "synced_documents_count": 3,
  "mahavault_synced": ["RATION_CARD", "INCOME_CERTIFICATE_FORM16", "DOMICILE_CERTIFIC
}
```


3. Dispatch Aadhaar OTP

`POST /api/v1/auth/aadhaar/send-otp`

```
{
  "status": "OTP_SENT",
  "txn_id": "UIDAI-TXN-881920",
  "masked_aadhaar": "XXXX-XXXX-9012",
  "resend_available_in_seconds": 45
}
```

4. Verify Aadhaar OTP & Complete e-KYC

`POST /api/v1/auth/aadhaar/verify-otp`

```
{
  "access_token": "eyJhbGciOi ... ",
  "token_type": "bearer",
  "user_id": "CIT-MH-88190",
  "full_name_en": "Rahul Suresh Patil",
  "masked_aadhaar": "XXXX-XXXX-9012",
  "district": "Pune",
  "ekyc_status": "VERIFIED_UIDAI_AADHAAR"
}
```